

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include <sys/socket.h>
5  #include <netinet/in.h>
6  #include <arpa/inet.h>
7  #include <unistd.h>
8  #include <errno.h>
9
10 #define BUF_SIZE 128
11
12 void err_exit(const char *err){
13     perror(err);
14     exit(1);
15 }
16
17 int main(int argc, char *argv[])
18 {
19     int sockfd = 0, msg_count = 0;
20     uint16_t port;
21     char buffer[BUF_SIZE] = {0,};
22     struct sockaddr_in server_addr;
23     if(argc != 3){
24         printf("\n Usage: %s server_ip port\n", argv[0]);
25         return 1;
26     }
27     port = atoi(argv[2]);
28     if((sockfd = socket(AF_INET, SOCK_STREAM, 0)) < 0)
29         err_exit("socket error");
30     memset(&server_addr, 0, sizeof(server_addr));
31     server_addr.sin_family = AF_INET;
32     server_addr.sin_port = htons(port);
33     inet_aton(argv[1], (struct in_addr*)&server_addr.sin_addr);
34     if(connect(sockfd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0)
35         err_exit("connect error");
36     while(1){
37         msg_count = recv(sockfd, buffer, BUF_SIZE, 0);
38         if(msg_count < 0) err_exit("recv error");
39         printf("(msg from server):%s\n", buffer);
40         while(1){
41             printf("message:");
42             fgets(buffer, BUF_SIZE, stdin);
43             if(strlen(buffer) >= 128 - 1)
44                 printf("message too large\n");
45             else
46                 break;
47         }
48         buffer[strlen(buffer) - 1] = '\0';
49         msg_count = send(sockfd, buffer, strlen(buffer) + 1, 0);
50         if(msg_count < 0)err_exit("send error");
51     }
52     shutdown(sockfd, SHUT_RDWR);
53     return 0;
54 }
```